

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: DIGITAL VLSI TESTING (01CT0629)	Aim: Detection of both SA0 and SA1 faults on a VLSI circuit for 3-bit EVEN parity generator circuit.	
Experiment No: 3	Date:04/03/2026	Enrollment No:92301733056

Aim: To analyze and detect stuck-at faults and to determine the output of the 3-bit even parity generator by using 2-input Ex-OR gates.

Simulation Platform: Virtual Lab

<https://vlsi-nitk.vlabs.ac.in/exp/3-bit-even-parity-generator-circuit/index.html>

Theory:

A **Parity Generator** is a combinational logic circuit that generates the parity bit in the transmitter. A parity generator adds an extra bit to the signal that is being transmitted, so the errors can be recognized and rectified by the machine to some extent.

Parity generators are classified into two types:

1. Even Parity Generator
2. Odd Parity Generator

Even Parity Generator:

A 3-bit even parity generator takes 3 inputs and adds an extra bit to the data, which is known as the parity bit. Then the data is transferred as 4 bits and the extra bit is the parity bit. In the even parity generator, it generates the bit such that the number of 1's in the input transferred (including the parity bit) is even. The circuit keeps track of the number of 1's in the data, if the data has the odd number of 1's, then the circuit detects an error.

Example: Let A, B, C be the inputs and P be the parity bit.

Let A=1, B=1, C=0

Now, the parity bit would be 0 to make the number of 1's even.

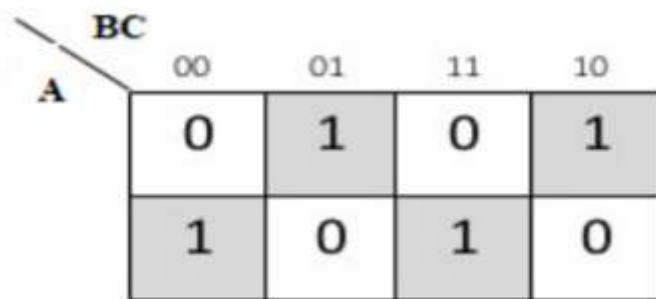
So P=0

Similarly,

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Table-1: Table For Even Bit Parity Generator

A	B	C	P(Parity bit)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	1	1
1	1	0	0



		BC			
		00	01	11	10
A	0	0	1	0	1
	1	1	0	1	0

Figure-1: K-map for the Even Bit Parity Generator

$$P = AB'C' + A'B'C + ABC + A'BC$$

$$P = A(B \odot C) + A'(B \oplus C)$$

$$P = A(\oplus)B(\oplus)C$$

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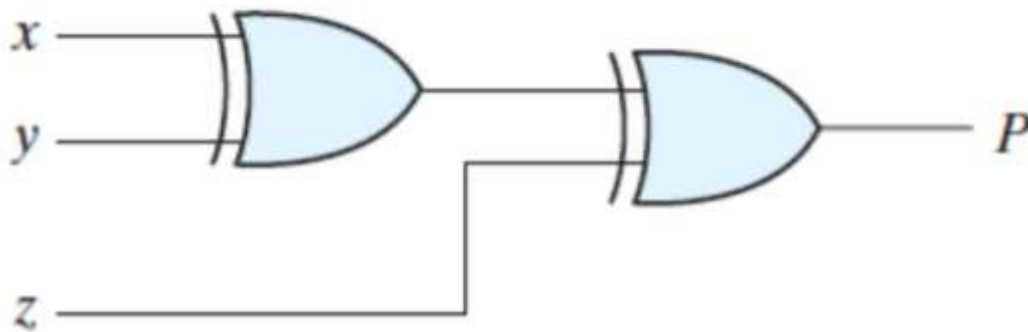


Figure-2: Circuit diagram of Even parity generator

Note: With this parity generator, the errors in the data transmission cannot be detected to a complete extent. It can only detect the error when only one of the bits changes. If two bits get changed, then the system cannot detect the error. Since the number of 1's being even or odd will remain the same.

Stuck-at Fault:

A stuck-at fault is a particular fault model used by fault simulators and automatic test pattern generation (ATPG) tools to mimic a manufacturing defect within an integrated circuit. Individual signals and pins are assumed to be stuck at Logical '1', '0' and 'X'.

Stuck-at faults are classified into two types: 1. SA0 fault 2. SA1 fault

SA0 Fault:

When a signal, or gate output, is stuck at 0 value, independent of the inputs to the circuit, then the signal is said to be "stuck at 0 fault" or "SA0 fault".

SA1 Fault:

When a signal, or gate output, is stuck at 1 value, independent of the inputs to the circuit, then the signal is said to be "stuck at 1 fault" or "SA1 fault". Output for the even parity bit when there is a stuck-at fault.

Example: Suppose there is a stuck-at 0 fault in wire A.

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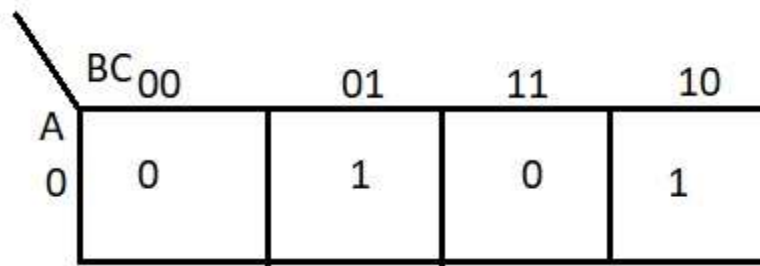


Figure-3: K-map for parity bit

$$P = A'B'C + A'BC'$$

$$P = A'(B \oplus C) \text{ \{A' is 1 since A has SA0\}}$$

$$P = B \oplus C$$

Table-2: Table For Even Parity Bit with Stuck-at Fault

X	Y	Z	$X \oplus Y$	P
X	Y	Z	$X \oplus Y$	$X \oplus Y \oplus Z$
SA0	Y	Z	Y	$Y \oplus Z$
SA1	Y	Z	Y'	$Y' \oplus Z$
X	SA0	Z	X	$X \oplus Z$
X	SA1	Z	X'	$X' \oplus Z$
X	Y	SA0	$X \oplus Y$	$X \oplus Y$
X	Y	SA1	$X \oplus Y$	$(X \oplus Y)'$
X	Y	Z	SA0	Z
X	Y	Z	SA1	Z'

Simulation:

Paste your simulation result screen shot here:

Simulation 1

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DETECTION OF BOTH SA0 AND SA1 FAULTS ON A VLSI CIRCUIT FOR 3-BIT EVEN PARITY GENERATOR CIRCUIT



Simulation



Add

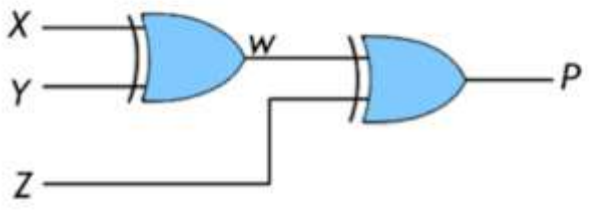


Print



Refresh

Simulation



Equation of fault detected = z

Fault	Wire	X	Y	Z
SA0	w	1	0	0

Simulation 2

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Truth Table:

Input			w	P	P'
x	y	z			
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	0	0	0
1	1	1	0	1	1
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	1	0
1	0	1	1	0	1
1	1	1	1	1	1

Conclusion:

In this experiment, we analyzed the detection of **Stuck-at-0 (SA0)** and **Stuck-at-1 (SA1)** faults in a 3-bit even parity generator circuit. By applying appropriate input test vectors, we compared the normal output with the faulty output to identify fault conditions. It was observed that specific input combinations are required to activate and propagate SA0 and SA1 faults to the output. The parity generator, which uses XOR logic, is sensitive to single-line faults, making fault detection easier through parity checking.

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The experiment demonstrated the importance of systematic test pattern generation in VLSI testing. Overall, the analysis improved understanding of fault modeling and reliability in digital circuit design.

Post Lab Exercise

1. How many XOR gates are required to construct an even parity generator for 3 input bits?

- (A) 2
- (B) 1
- (C) 3
- (D) 4

2. Parity Generator is a combinational logic circuit that generates the parity bit in the transmitter.

- (A) True
- (B) False

3. If the signal 101 is transmitted but received as 011, which of the following statements is correct?

- (A) Error cannot be detected
- (B) Error can be detected using 3-bit odd parity generator
- (C) Error can be detected using 3-bit even parity generator
- (D) Both b and c are correct

4. What is the odd parity output for the decimal number 8?

- (A) 1
- (B) 110
- (C) 1000

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(D) 0

5. Which gates are used to construct an odd parity generator?

- (A) Ex-OR & NOT gates
- (B) Ex-OR & AND gates
- (C) Ex-OR gates
- (D) NAND & NOR gates

6. If a Stuck-At-1 (SA1) fault occurs at input wire A, what will be the resulting Boolean function for the parity bit?

- (A) $B' \oplus C$
- (B) $A' \oplus B$
- (C) C
- (D) $A \oplus B$

7. In the case of 3-bit even even parity generator, when all the inputs are high and 'A' is stuck at _____ output at _____ .

- (A) 0 1
- (B) 1 0
- (C) 0 0
- (D) None of the above

8. How many types of parity bits are commonly used in digital systems?

- (A) 4
- (B) 2
- (C) 5
- (D) 3

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9. Which of the following is the correct Boolean expression for an even parity generator using three input variables?

- (A) $A \oplus B \oplus C'$
- (B) $A \oplus B + B \oplus C + A \oplus C$
- (C) $A \oplus B \oplus C$
- (D) $A + B + C$

10. For which of the following binary data inputs is the even parity bit equal to '1'?

- (A) 101
- (B) 110
- (C) 1011
- (D) 0101